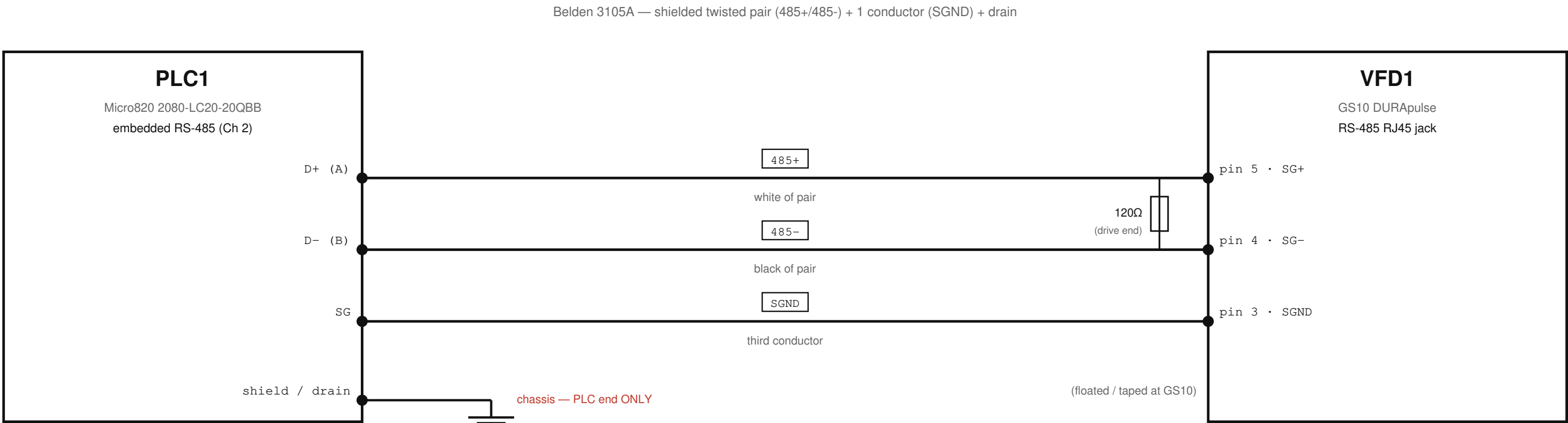


E-007 RS-485 / MODBUS RTU COMMUNICATION

Conv_Simple / CV-101 (garage conveyor) — Micro820 (master) ↔ GS10 (node 1) · Modbus RTU only



link up ⇒ vfd_comm_ok = TRUE ; read 0x2103 (FC03) for nonzero freq

CONNECTION TABLE

Src dev	Src terminal	Dst dev	Dst terminal/pin	Cable / conductor	Wire	Evidence	Notes
PLC1	D+ (A)	VFD1	RJ45 pin 5 / SG+	Belden 3105A pair · white of pair	485+	verified	Polarity: if silent/garbled, swap 485+/485- first.
PLC1	D- (B)	VFD1	RJ45 pin 4 / SG-	Belden 3105A pair · black of pair	485-	verified	
PLC1	SG	VFD1	RJ45 pin 3 / SGND	3rd conductor · third conductor	SGND	verified	Dedicated wire, NOT the shield.
PLC1	shield / chassis	VFD1	(floated)	tinned drain · drain	SH	field_verify	Land drain at PLC end ONLY; tape/float at GS10 end.

CCW SERIAL PORT: Modbus RTU, Master · 9600 · 8N1 (P09.04 = 12) · Channel 2 · Node 1 (P09.00 = 1)

Older GS10_Integration_Guide.md said 8N2 (P09.04=13) - SUPERSEDED by bench-verified 8N1.

CORRECTED from May-16 draft: Channel 0 → 2 · SGND pin 1/8 → pin 3 · 8N2 → 8N1. Do not copy the old values.

TROUBLESHOOTING (this circuit)

- Nothing after config -> swap 485+/485- (A/B polarity; vendors disagree).
- CRC errors -> usually polarity. Complete silence -> usually baud mismatch (confirm 9600 8N1).
- GS10 CE10 / F30 comm-timeout -> no Modbus packets; check wiring, baud, P09.03 timeout.
- 120 ohm termination across SG+/SG- at the GS10 end for long runs (bench <2 m usually fine).
- SGND is a dedicated conductor, not the shield; missing common -> intermittent corruption.

- VERIFIED (recovered CommsToVFD + Beginner_Verify p48 + Prog_init v2.1)
- FIELD VERIFY (exact PLC chassis ground point; cable P/N if not Belden 3105A)

SOURCES: CCW/MIRA_PLC/docs/instructions/Conv_Simple_CommsToVFD.pdf §2 (recovered style + wiring)
CCW/MIRA_PLC/docs/instructions/Conv_Simple_GS10_Beginner_Verify_V2.pdf p48 (corrected pins/channel)
plc/Prog_init_ConvSimple_v2.1.st (Channel 2, node 1, 0x2103)
plc/GS10_Integration_Guide.md §3 (RJ45 pinout, register map)

RS-485 / MODBUS RTU Conv_Simple / CV-101 (garage conveyor)	SHEET E-007
MIRA / FactoryLM recovers MIRA-WI-001 / Conv_Simple_CommsToVFD §2	REV A 2026-07-037 of 9